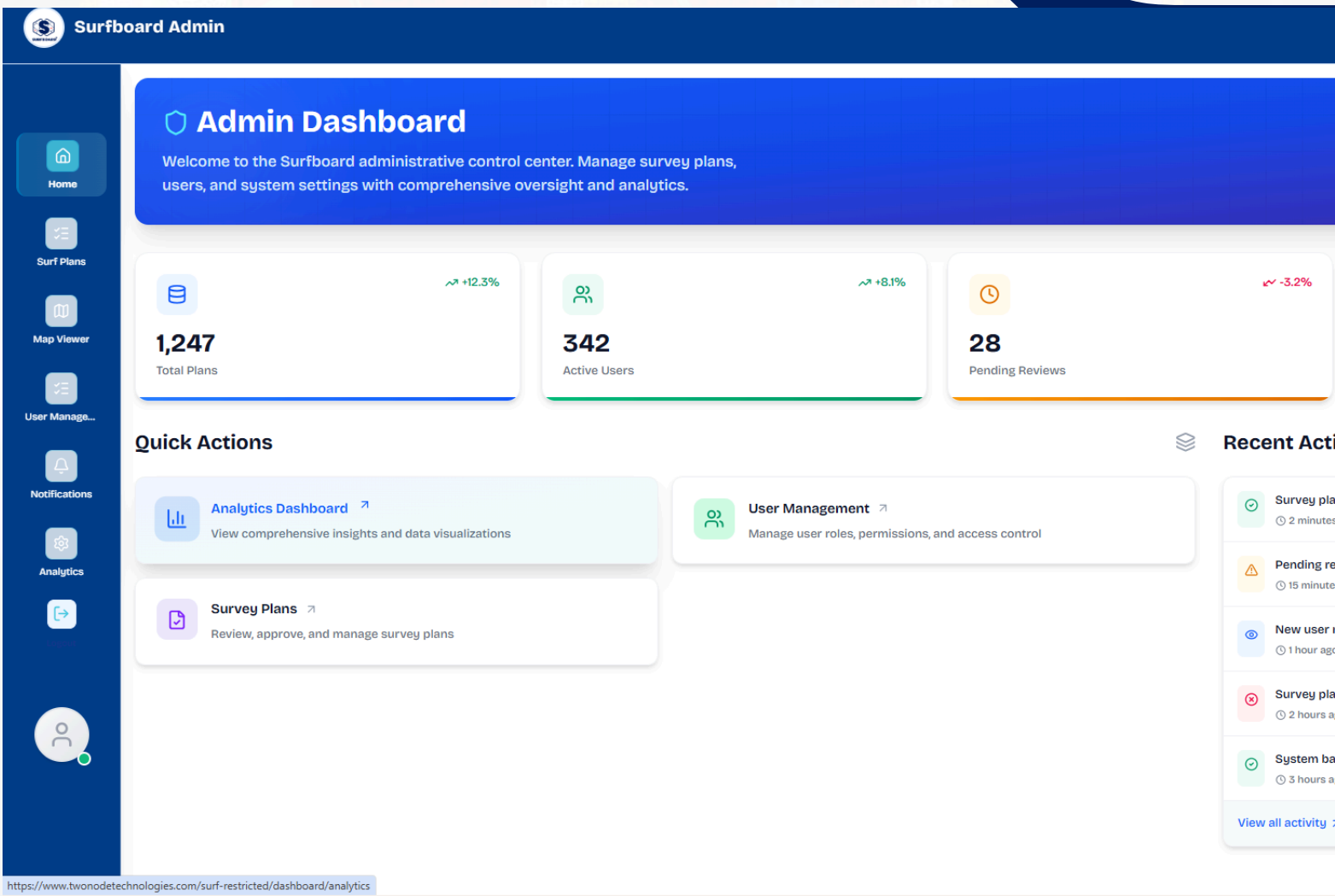
TwoNode Technologies

PRODUCT BRIEF II

SURFBOARD®



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1.0 Transaction Completion Logic A survey plan is considered fully completed on Surfboard when:

- The field certificate_ready is set to “Yes”, and
- The corresponding certificate file is attached.

2.0 External Process – MDA Certificate of deposit Generation

To generate this certificate, the surveyor must submit a document called “record copy” to an MDA:
Office of the State Surveyor General (OSSG).

The MDA will now use surfboard under different roles

3.0 Data Currently Available on Surfboard

Surfboard already stores all the information needed by OSSG to process a certificate, including:

- Survey Title
- Parcel Area (Size)
- LGA
- Ward/LCDA
- Deeded Area
- Pillar Numbers and Coordinates
- Record Copy (PDF or Image) – newly required but already in our schema, uploaded by the surveyor
- Locality - **New Field**
- Possible Charting Remarks - **New Field**

4.0 Records that can be submitted

Status Fields Reminder
There are two separate status fields in the schema:

1. status → (default: Pending)
2. redCopyStatus → (Pending, Submitted, Queried)

Only survey plans with:
✓ status = Pending
✓ redCopyStatus = Pending
…can be submitted for Certificate of Deposit.

STEP 1 – Batch Submission of Survey Plans

Users can select multiple survey plans to submit as a batch.

Let's call this Batch Submission.

When they click Submit for Certificate of Deposit, the system must do the following:

(A) Perform Spatial Intersection (PostGIS):

Intersect each survey plan with the PostGIS dataset:

LASG_charting_interpretation

- Field to read → interpretation
- Result should populate the field → possibleChartingRemarks

This similar to what we did with Nigeria_ward_admin_boundaries

(B) Compute Certificate of Deposit Fees

Fee = (Rate per LGA * Area)

Configuration:

- Each LGA has a fee per square meter (editable by Surfboard Super Admin)
  - Example: Eti-Osa = ₦10/m²
  - Survey size = 500 m²
  - Fee = 10 × 500 = ₦5,000

Add system charges = 10%

Total per survey = Fee + Charges

Example:

- 10 surveys × ₦5,500 each = ₦55,000

(C) Batch Code Assignment

Every batch submission is assigned a unique batch code.

All survey records inside that batch inherit the batch code.

(D) Payment Flow (Two-Fold)

Fold 1 – User Payment via Paystack

- User pays the total batch amount
- After successful payment:
  - All Surfboard Admins receive email + notification

Fold 2 – Surfboard Admin Payment to Lagos State Government (Manual)

- Surfboard Admin manually remit to LASG
- Uploads the LASG receipt to the batch
- Upon uploading receipt:
  - Roles - Records I, Records II, Accounts, Charting, SURCON all receive email + notification

(E) Update Record-Level Statuses

After submission:

- redCopyStatus: Pending → Submitted
- status: remains Pending

ROLE-BASED ACTION WORKFLOWS

1. Accounts Admin

Sees:

Only Awaiting Accounts Approval batch transactions with LASG receipt attachment.

Task: Approves or disapproves the payment. (disapproval should send a mail only to surfboard admins)

Next: Batch becomes invisible to Accounts Admin.

2. Charting Admin

Sees:

All submitted survey plans with:

- Account status
- Charting-related fields

Task: Edits or confirms possibleChartingRemarks.

Next: Charting status changes to Approved.

3. SURCON Admin

Sees:

All submitted survey plans with:

- Account status
- Charting status

Task: Confirms pillar numbers:

- Approve OR
- Query (with comment)
    - Automatically email + notification sent to the user indicating query stage.

Next: Record becomes invisible to SURCON upon Approval.

4. Records I

Sees:

All submitted survey plans with:

- Account status
- Charting status
- SURCON status

Task: Confirms survey plan Record Copy (attachment)

- Approve OR
- Query (with comment → user notified automatically)

Next: Record becomes invisible to Records I upon approval.

5. Records II

Sees:

All submitted survey plans with:

- Account status
- Charting status
- SURCON status
- Records I status

Task: Prepares to collect hard copy by verifying:

- Online uploaded copy
- Physical copy submitted

Final notes each role duties can occur simultaneously but for Records II all the duties of other roles must have been Approved.

Next:

After approval:

- Fields listed in **3.0** should be sent to the existing Certificate of Deposit API (OSSG)
- Survey plans become invisible
- Email sent to user should contain:
  - Batch number
  - List of plan numbers
  - Survey titles
  - Facilitator
  - Notification that plans have been successfully lodged
- status of each survey changes to: → Lodged
- redCopyStatus of each survey changes to: → Approved
- (Later, we will add a process to retrieve certificate files from the LASG portal using the Plan No. from this url “http://ossgdeposit.lagosstate.gov.ng/survey/check.php” using plan no as input.)